

Assignment-1

Q.1 Write a C++ program to find out maximum number from the given Array

```
#include <iostream.h>
#include <conio.h>
#include <iomanip.h>
class Sort
{
    private:      int value[100];
    public :      Sort(int );
                  int  maxvalue(int );
                  ~Sort() { cout << " Destructor Called..."<<endl;}
};
Sort:: Sort(int n)
{
    cout << "Constructor Called..."<<endl;
    for (int i=0;i<n;i++)
    {
        cout <<"Enter ["<<i<<"]= " ;
        cin  >> value[i];
    }
}
int Sort::maxvalue(int n)
{
    int max=value[0];
    for(int i=1;i<n;i++)
    {
        if (max < value[i])
            max = value[i];
    }
    return max;
}
void main()
{
    int n;
    clrscr();
    cout<< "How many values are there? : ";      cin>>n;
    Sort s(n);
    cout <<"Max value is : " << s.maxvalue(n);
    getch();
}
// O U T P U T
```

Q.2 Write a C++ program to find out minimum number from the given Array

➤ Structure of C++ program is given below:

➤ Class Minimum

```
{  
    Private: float value [100];  
    Public : Minimum()  
            int minimumvalue (int n);  
            ~Minimum();  
};
```

➤ You are instructed to write appropriate function code for the above class.

➤ Write appropriate main()

Q.3 Write a C++ program to find out whether given string is palindrome or not

➤ Structure of C++ program is given below:

➤ Class Palindrome

```
{  
    Private :   char str[50];  
               int flag;  
    public :   Paylindrome();  
               void find_palindrome();  
               void display();  
               ~Palindrome();  
};
```

➤ Get the string with the use of Palindrome() constructor member function

➤ Write the logic in find_palindrome() normal member function and set flag=0 if given string is not palindrome or flag=1 if given string is palindrome

➤ Based on flag value display appropriate message with the use of display()

➤ Write the code for ~Palindrome() destructor member function

➤ Write appropriate main()

- Input: madam ➔ Palindrome
- Malayalam ➔ Palindrome

Q.4 Write a C++ program to sort given array

- Structure of C++ program is given below:
- Class Sort_value

```

{
    Private :   int a[50];
    public :   Sort_value (int n);
               void sort (int n);
               void display(int n);
               ~Sort_value();
};

```

- Get the array value based on parameter n with the use of constructor member function.
- Write the logic in Sort(int n) member function
- Display sorted value with the use of display (n)
- Write the code for ~Sort_value()
- Write appropriate main()

Q.5 Write a C++ program to find out given array is sorted or not

- Structure of C++ program is given below:
- Class Array_5

```

{
    Private :   int a[50];
               char answer;
    public :   Array_5 (int n);
               void sort_decide (int n);
               void display();
               ~Array_5();
};

```

- Get the array value based on parameter n with the use of constructor member function.
- Write a logic in sort_decide(int n) and set answer value as follows

Answer =A ➔ Array is in Ascending Order

Answer = B ➔ Array is in Descending order

Answer = C ➔ Array is in no order

Answer = D ➔ All elements are equal

➤ Based on above answer appropriate message will be display by display() member function.

➤ Write the code for ~Array_5()

➤ Write appropriate main()

➤ Example

9	12	14	22	26	39	45	48
a[0]	a[1]	a[2]	a[3]	a[4]	a[5]	a[6]	a[7]

Answer =A ➔ Array is in Ascending Order

67	55	31	26	24	20	15	14
a[0]	a[1]	a[2]	a[3]	a[4]	a[5]	a[6]	a[7]

Answer = B ➔ Array is in Descending order

25	36	56	66	32	77	88	99
a[0]	a[1]	a[2]	a[3]	a[4]	a[5]	a[6]	a[7]

Answer = C ➔ Array is in no order

55	55	55	55	55	55	55	55
a[0]	a[1]	a[2]	a[3]	a[4]	a[5]	a[6]	a[7]

Answer = D ➔ All elements are equal